

For a phase margin of $PM = 55^\circ$, we need

$$\angle A(j\omega'_t) = -180^\circ + PM = -125^\circ \quad (6.55)$$

Setting (6.54) equal to (6.55) and solving for ω'_t/ω_2 results in

$$\begin{aligned} \tan^{-1}(\omega'_t/\omega_2) &= 35^\circ \\ \Rightarrow \omega'_t &= 2.2 \times 10^8 \text{ rad/s} = 2\pi \times 35 \text{ MHz} \end{aligned} \quad (6.56)$$

Next, setting $|A(j\omega'_t)| = 1$ and again using (6.53) results in

$$\begin{aligned} \frac{A_0}{(\omega'_t/\omega_{p1})\sqrt{1 + (\omega'_t/\omega_2)^2}} &= 1 \\ \Rightarrow \omega_{p1} &= \frac{\omega'_t\sqrt{1 + (\omega'_t/\omega_2)^2}}{A_0} = 2\pi \times 4.28 \text{ kHz} \end{aligned} \quad (6.57)$$

(b) First, we set

$$\omega_z = 1.7\omega'_t = 2\pi \times 59.5 \text{ MHz} \quad (6.58)$$

To find the new unity-gain frequency, setting $|A(j\omega_t)| = 1$ in (6.53) now gives

$$\begin{aligned} \frac{A_0\sqrt{1 + (\omega_t/\omega_z)^2}}{(\omega_t/\omega_{p1})\sqrt{1 + (\omega_t/\omega_2)^2}} &= 1 \\ \Rightarrow \omega_t &= \frac{A_0\omega_{p1}\sqrt{1 + (\omega_t/\omega_z)^2}}{\sqrt{1 + (\omega_t/\omega_2)^2}} \end{aligned} \quad (6.59)$$

This equation can be solved for the new unity-gain frequency. Both sides can be squared, resulting in a quadratic equation in ω_t , which can be solved exactly. Alternatively, the value for ω'_t found in part (a) can be used as an initial guess, and (6.59) can be solved iteratively. After four iterations, one finds that $\omega_t = 2\pi \times 39.8 \text{ MHz}$. This unity-gain frequency is a 14 percent increase over that found in part (a). We can solve for the new phase shift by using

$$\angle A(j\omega_t) = -90^\circ + \tan^{-1}(\omega_t/\omega_z) - \tan^{-1}(\omega_t/\omega_2) = -95^\circ \quad (6.60)$$

Note that this phase value gives a phase margin of 85° , which is a 30° improvement. In practice, the improvement may not be this great due to additional high-frequency poles and zeros (which have been ignored here). Regardless of this degradation, the improvement from using lead compensation is substantial.

6.2.3 Making Compensation Independent of Process and Temperature

This section shows how lead compensation can be made process and temperature insensitive. Repeating equations (6.9) and (6.20) here, we have

$$\omega_t = \frac{g_{m1}}{C_C} \quad (6.61)$$

and

$$\omega_{p2} \cong \frac{g_{m7}}{C_1 + C_2} \quad (6.62)$$

$$\omega_z = \frac{-1}{C_C(1/g_{m7} - R_C)} \quad (6.63)$$

$$\omega_z = \frac{-1}{C_c(1/g_{m7} - R_c)} \quad (6.63)$$

It turns out that R_C can be made proportional to $1/g_{m7}$ as long as R_C is realized by a transistor in the triode region that has an effective gate-source voltage proportional to that of Q_7 . To see this result, recall that R_C is actually realized by Q_6 , and therefore we have

$$R_C = r_{ds9} = \frac{1}{\mu_n C_{ox} (W/L)_9 V_{eff9}} \quad (6.64)$$

[illegible]

$$g_{m7} = \mu_n C_{ox}(W/L)_7 V_{eff7} \quad (6.65)$$

$$R_{cg_{m7}} = \frac{(W/L)_7 V_{eff7}}{(W/L)_9 V_{eff9}} \quad (6.66)$$

Therefore, all that remains is to ensure that $V_{\text{eff9}}/V_{\text{eff7}}$ is independent of process and temperature variations since clearly the remaining terms depend only on a geometric relationship. The ratio $V_{\text{eff9}}/V_{\text{eff7}}$ can be made constant by deriving V_{GS9} from the same biasing circuit used to derive V_{GS7} . Specifically, consider the circuit shown in Fig. 6.11 which consists of a bias stage, the second stage, and the compensation network of the two-stage opamp. First we must make $V_a = V_b$, which is possible by making $V_{\text{eff13}} = V_{\text{eff7}}$. These two effective gate-source voltages can be made equal by taking

$$\sqrt{\frac{2I_{D7}}{\mu_n C_{ox}(W/L)_7}} = \sqrt{\frac{2I_{D13}}{\mu_n C_{ox}(W/L)_{13}}} \quad (6.67)$$

Squaring and simplifying, we see that the following condition must be satisfied:

$$\frac{I_{D7}}{I_{D13}} = \frac{(W/L)_7}{(W/L)_{13}} \quad (6.68)$$

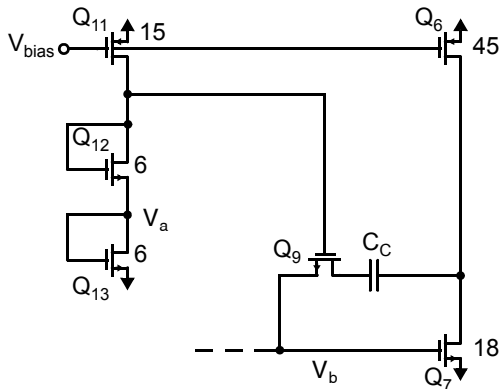


Fig. 6.11 The bias circuit, second stage, and compensation circuit of the two-stage opamp.

11. The ratio μ_n/μ_p is relatively constant for a given process but often varies considerably from process to process (say, from 0.6- μm to 0.4- μm CMOS processes).

However, the ratio I_{D7}/I_{D13} is set from the current mirror pair Q_6, Q_{11} , resulting in

$$\frac{I_{D7}}{I_{D13}} = \frac{(W/L)_6}{(W/L)_{11}} \quad (6.69)$$

Thus, to make $V_{\text{eff}13} = V_{\text{eff}7}$, from (6.68) and (6.69), we need to satisfy the following relationship:

$$\frac{(W/L)_6}{(W/L)_7} = \frac{(W/L)_{11}}{(W/L)_{13}} \quad (6.70)$$

Assuming this condition is satisfied, we have $V_{\text{eff}13} = V_{\text{eff}7}$, and therefore we also have $V_a = V_b$. Since the gates of Q_{12} and Q_9 are connected and their source voltages are the same, we also have $V_{\text{eff}12} = V_{\text{eff}9}$. Next, using these gate-source relationships and noting that $I_{D12} = I_{D13}$, we can write

$$\frac{V_{\text{eff}7}}{V_{\text{eff}16}} = \frac{V_{\text{eff}13}}{V_{\text{eff}12}} = \frac{\sqrt{\frac{2I_{D13}}{\mu_n C_{\text{ox}}(W/L)_{13}}}}{\sqrt{\frac{2I_{D12}}{\mu_n C_{\text{ox}}(W/L)_{12}}}} = \sqrt{\frac{(W/L)_{12}}{(W/L)_{13}}} \quad (6.71)$$

Finally, substituting (6.71) into (6.66), we have the product $R_C g_{m7}$, given by

$$R_C g_{m7} = \frac{(W/L)_7}{(W/L)_9} \sqrt{\frac{(W/L)_{12}}{(W/L)_{13}}} \quad (6.72)$$

which is only dependent on geometry and not on processing or temperature variations.

As a result, we have guaranteed that the drain-source resistance of a transistor in the triode region is inversely matched to the transconductance of a different transistor. This relationship can be very useful for many other applications as well. Indeed, in Chapter 7 we will see that it's quite simple to make all of the transconductances of transistors in a microcircuit match the conductance of a single off-chip resistor. *This approach results in the possibility of on-chip "resistors," realized by using triode-region transistors that are accurately ratioed with respect to a single off-chip resistor.* This relationship can be very useful in modern circuit design.

6.3 ADVANCED CURRENT MIRRORS

6.3.1 Wide-Swing Current Mirrors

As newer technologies with shorter channel lengths are used, it becomes more difficult to achieve reasonable opamp gains due to transistor output-impedance degradation caused by short-channel effects. As a result, designers are often forced to use cascode current mirrors. Unfortunately, the use of conventional cascode current mirrors limits the signal swings available, which may not be tolerated in certain applications. Fortunately, circuits exist that do not limit the signal swings as much as the current mirrors discussed in Chapter 3. One such circuit is shown in Fig. 6.12 and is often called the "wide-swing cascode current mirror" [Sooch, 1985; Babanezhad, 1987].

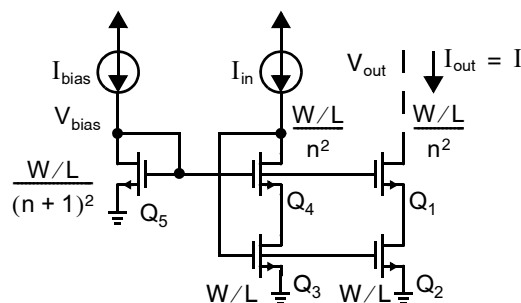


Fig. 6.12 The wide-swing cascode current mirror.